8 A typical three-phase generator in turbines use for power generation consists of a rotating electromagnet surrounded by three sets of stationary wire coils AA', BB', CC', with angles of 120˚ between them, wrapped around metal cores to generate electricity, as shown in Fig. 8.1.

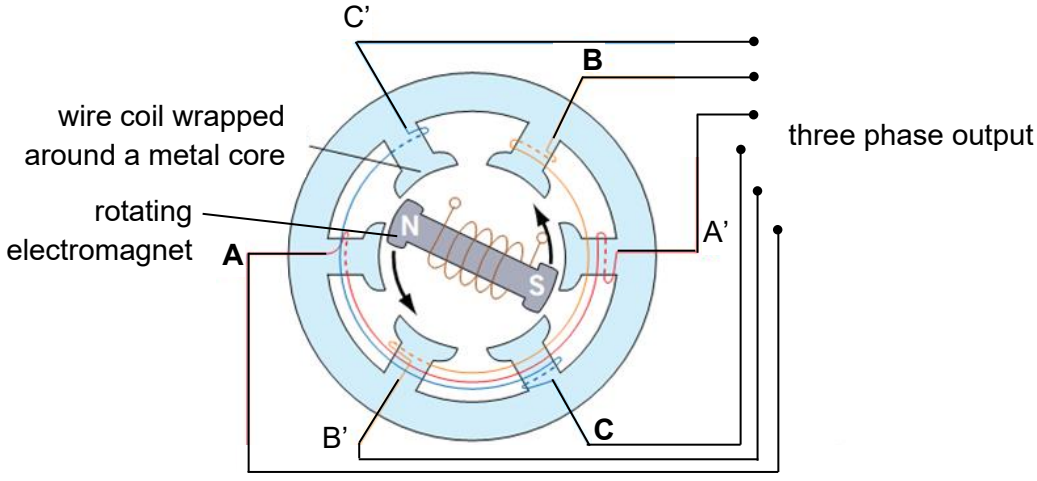

Fig. 8.1

Each set of coils (AA', BB', CC') is a solenoid that consists of wires tightly wound around a rigid metal core as shown in Fig. 8.2.

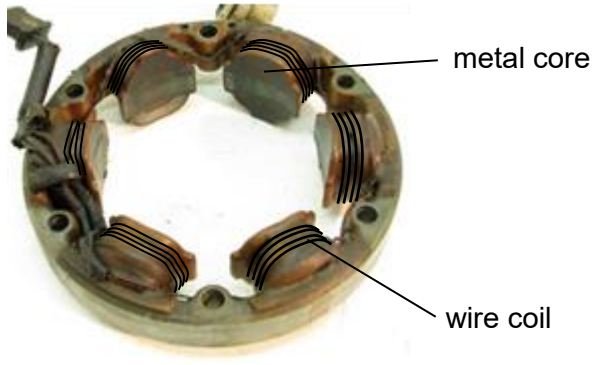

Fig. 8.2

The electromagnet shown in Fig. 8.1 can rotate about the axis perpendicular to the plane of the page, and is connected to turbine blades that rotate the electromagnet at a constant rate such that electricity can be generated in the coils.

(a) State Faraday's law of electromagnetic induction.

……...

……...

…….. [2]

(b) **(i)** Ignoring coils B and C, explain why the rotation of the electromagnet causes a sinusoidal e.m.f. to be induced in coil A.

.....

.....

.....

.....

..... [3]

(ii)

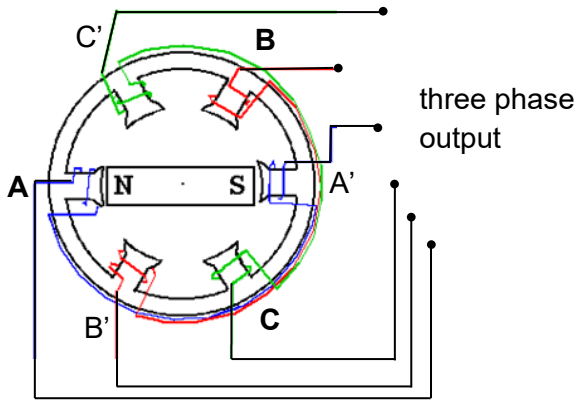

Fig. 8.3

At a particular instant, at time t = 0 s, the magnet is aligned with the set of coils labelled AA' as shown in Fig. 8.3.

Set AA' has a total of 500 turns with a diameter 0.10 m. The electromagnet generates an effective uniform magnetic flux density of 0.80 T at the position of the coils when its axis aligns with that of the coils.

Calculate the magnetic flux linkage of the coils at this instant. Give an appropriate unit.

magnetic flux linkage = ... [3]

(iii) The variation with time of magnetic flux linkage of the coil is as shown in Fig. 8.4. The flux linkage varies sinusoidally with time.

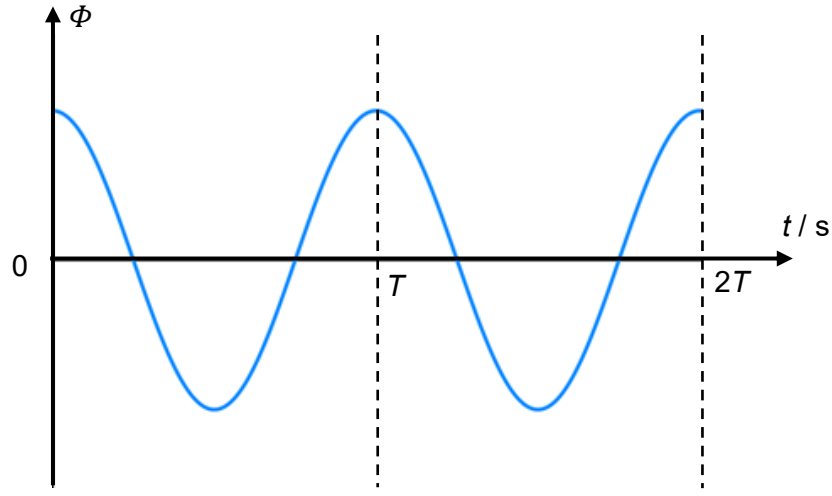

Fig. 8.4

When operating optimally, the magnet makes 1800 revolutions per minute.

Write an expression for the magnetic flux linkage Φ in the coil as a function of time t as shown in Fig. 8.4.

[1]

(iv) Hence, show that the maximum induced e.m.f. in the coil is 590 V.

[2]

(v) On Fig. 8.5, sketch the corresponding graph of the variation with time of induced e.m.f. in the coil AA'. Explain, with reference to Fig. 8.4, the position of maximum induced e.m.f. in your graph.

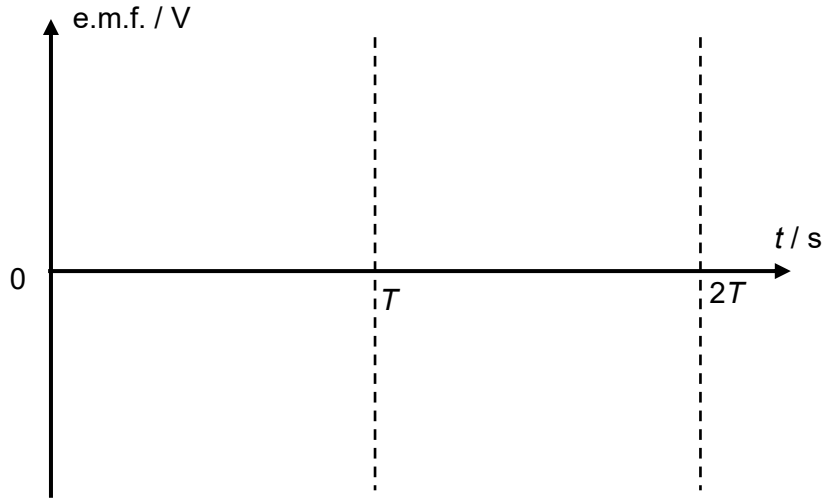

Fig. 8.5

.....

.....

.....

..... [3]

(c) Two additional sets of wire coils BB' and CC' complete the three-phase generator as shown in Fig. 8.3. As the electromagnet rotates anti-clockwise, the induced e.m.f. in coils AA', BB' and CC' are $\frac{2\pi}{3}$ rad out of phase with each another.

On Fig. 8.5, sketch additional graphs to show how the induced e.m.f. in coils BB' and CC' vary with time. Label your sketches clearly.

[2]

(d) The output of the generator is connected to the input of an ideal transformer as shown in Fig. 8.6.

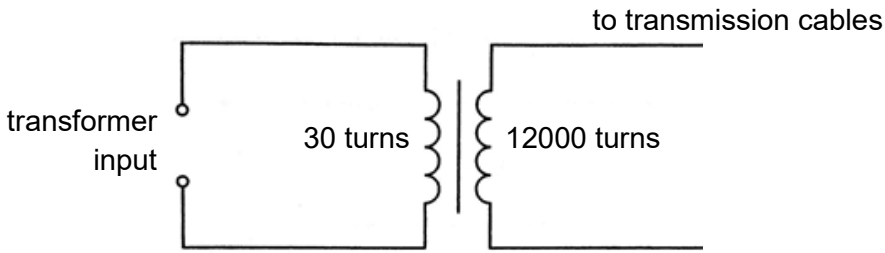

Fig. 8.6

The peak input current is measured to be 20 A. The transformer has 30 turns on its primary coil and 12000 turns on its secondary coil. The secondary coil circuit is connected to transmission cables used for power transmission to homes more than 20 km away.

(i) Explain what is meant by an *ideal* transformer.

.....

..... [1]

(ii) Determine the r.m.s. current in the secondary coil.

r.m.s. current = ... A [2]

(iii) Hence, or otherwise, suggest why the e.m.f. generated is stepped up to high voltages for long distance transmission.

.....

.....

..... [1]